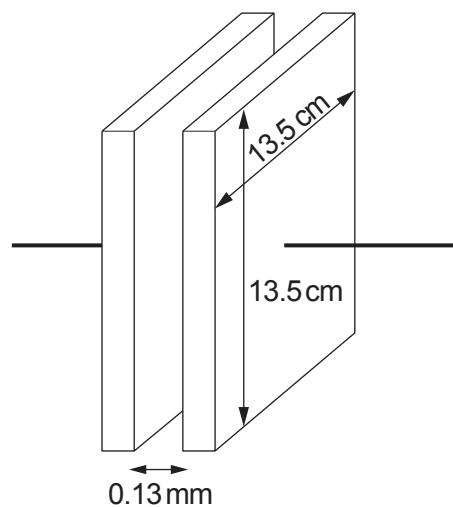


4. (a) Calculate the capacitance of the capacitor shown. [2]

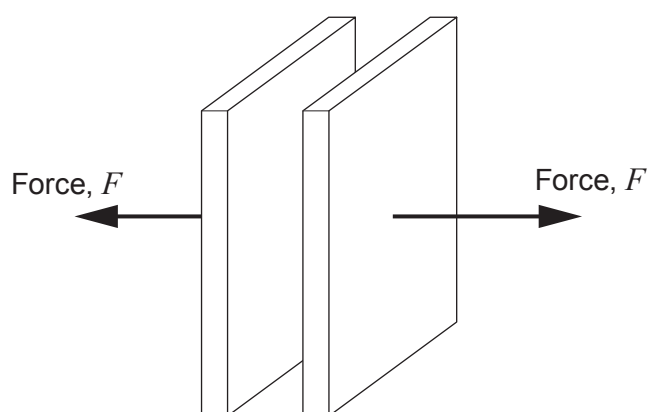


- (b) (i) Calculate the pd across the terminals of a 5.0 mF capacitor when it stores 1.00 J of energy. [2]



- (ii) In the space provided, draw a diagram to show how you would combine three 5.0 mF capacitors to produce a capacitance of 7.5 mF . [2]

- (c) The separation of the plates of a charged capacitor is increased by the application of a force, F . The capacitor is **isolated** so the charges on the plates remain unchanged.



- (i) State why a force must be exerted to separate the charged plates. [1]

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- (ii) Explain why the capacitor stores more **energy** when the separation of the plates is increased even though the charge remains constant. [2]

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- (iii) Show that the energy stored by the capacitor is given by:

$$\text{energy stored} = \frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$$

where Q is the charge stored, d is the separation of the plates, ϵ_0 is the permittivity of free space and A is the area of the plates. [2]

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- (iv) Bethan states that the force, F , required to separate the plates is given by:

$$F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$$

Determine whether Bethan is correct to arrive at this conclusion. [2]

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